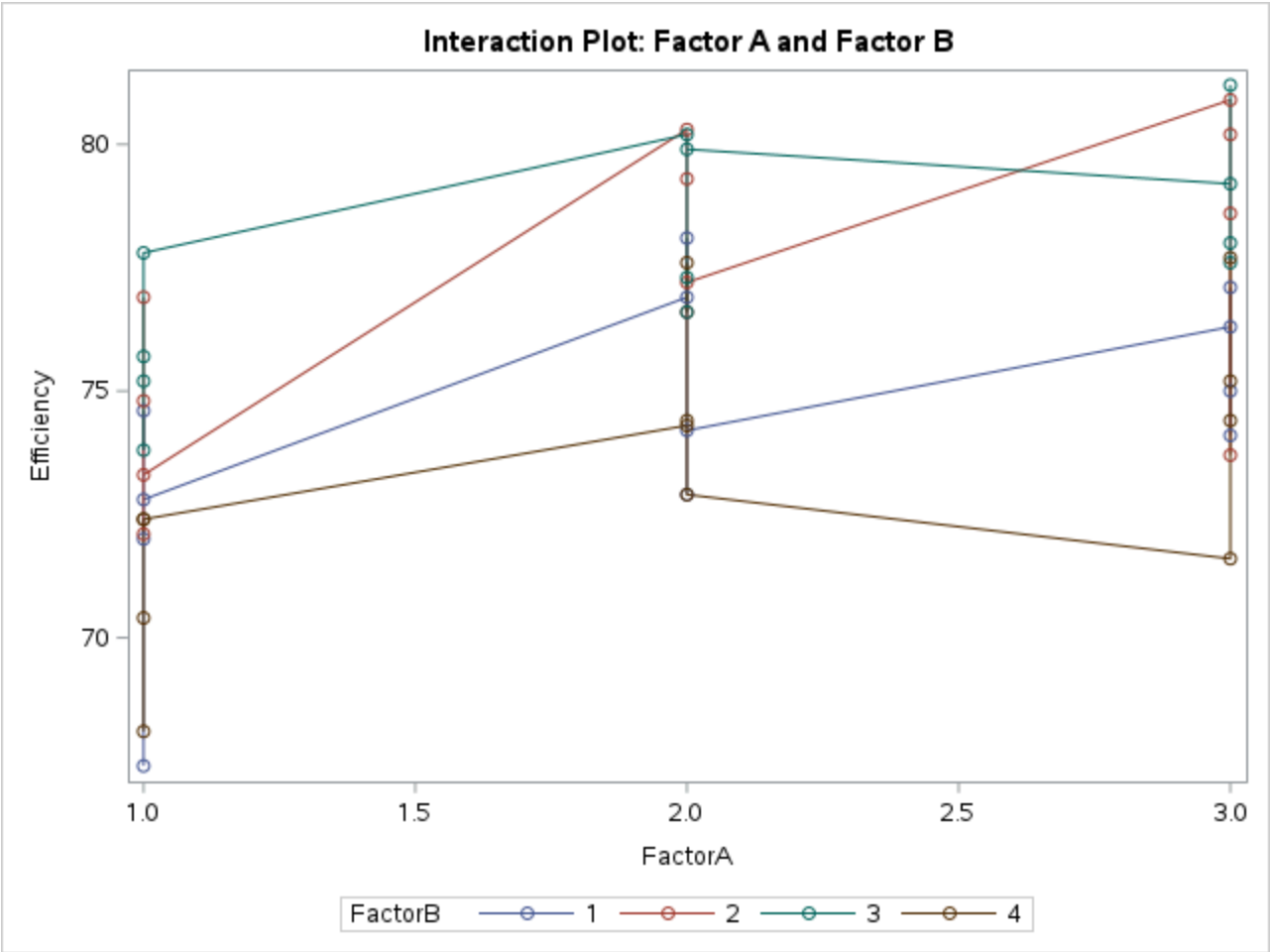




PROC GLM: Testing Interaction Term

The GLM Procedure

Class Level Information		
Class	Levels	Values
FactorA	3	1 2 3
FactorB	4	1 2 3 4

Number of Observations Read	48
Number of Observations Used	48

PROC GLM: Testing Interaction Term

The GLM Procedure

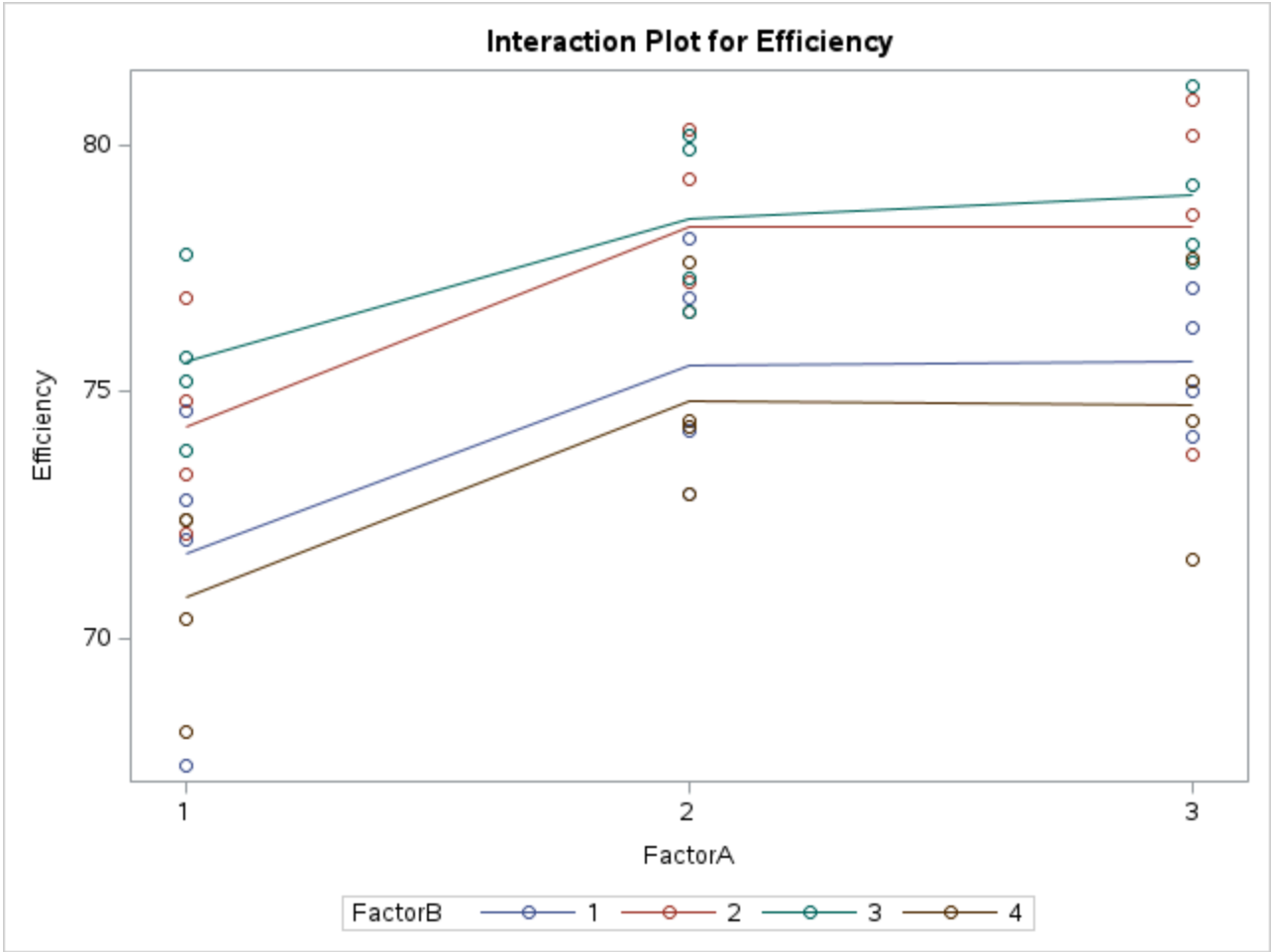
Dependent Variable: Efficiency

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	11	305.0916667	27.7356061	5.75	<.0001
Error	36	173.6250000	4.8229167		
Corrected Total	47	478.7166667			

R-Square	Coeff Var	Root MSE	Efficiency Mean
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R-Square	Coeff Var	Root MSE	Efficiency Mean
0.637312	2.904593	2.196114	75.60833

Source	DF	Type III SS	Mean Square	F Value	Pr > F
FactorA	2	150.3879167	75.1939583	15.59	<.0001
FactorB	3	152.8516667	50.9505556	10.56	<.0001
FactorA*FactorB	6	1.8520833	0.3086806	0.06	0.9988



PROC GLM: Testing Interaction Term

The GLM Procedure

Source	Type III Expected Mean Square
FactorA	Var(Error) + 4 Var(FactorA*FactorB) + Q(FactorA)
FactorB	Var(Error) + 4 Var(FactorA*FactorB) + 12 Var(FactorB)
FactorA*FactorB	Var(Error) + 4 Var(FactorA*FactorB)

PROC GLM: Testing Interaction Term

The GLM Procedure

Tests of Hypotheses for Mixed Model Analysis of Variance

Dependent Variable: Efficiency

Source	DF	Type III SS	Mean Square	F Value	Pr > F
Error: MS(FactorA*FactorB)					

Source	DF	Type III SS	Mean Square	F Value	Pr > F
FactorA	2	150.387917	75.193958	243.60	<.0001
FactorB	3	152.851667	50.950556	165.06	<.0001
Error	6	1.852083	0.308681		
Error: MS(FactorA*FactorB)					

Source	DF	Type III SS	Mean Square	F Value	Pr > F
FactorA*FactorB	6	1.852083	0.308681	0.06	0.9988
Error: MS(Error)	36	173.625000	4.822917		

PROC MIXED: REML Estimates for Variance Components

The Mixed Procedure

Model Information	
Data Set	WORK.TRAINING
Dependent Variable	Efficiency
Covariance Structure	Variance Components
Estimation Method	REML
Residual Variance Method	Profile
Fixed Effects SE Method	Model-Based
Degrees of Freedom Method	Satterthwaite

Class Level Information		
Class	Levels	Values
FactorA	3	1 2 3
FactorB	4	1 2 3 4

Dimensions	
Covariance Parameters	3
Columns in X	4
Columns in Z	16
Subjects	1
Max Obs per Subject	48

Number of Observations	
Number of Observations Read	48
Number of Observations Used	48
Number of Observations Not Used	0

Iteration History			
Iteration	Evaluations	-2 Res Log Like	Criterion
0	1	225.45311480	
1	2	207.86808217	0.00000079
2	1	207.86803230	0.00000000

Convergence criteria met.

Estimated G matrix is not positive definite.

Covariance Parameter Estimates	
Cov Parm	Estimate
FactorB	3.8974
FactorA*FactorB	0
Residual	4.1780

Fit Statistics	
-2 Res Log Likelihood	207.9
AIC (Smaller is Better)	211.9
AICC (Smaller is Better)	212.2
BIC (Smaller is Better)	210.6

Type 3 Tests of Fixed Effects				
Effect	Num DF	Den DF	F Value	Pr > F
FactorA	2	42	18.00	<.0001

PROC MIXED: ML Estimates for Variance Components

The Mixed Procedure

Model Information	
Data Set	WORK.TRAINING
Dependent Variable	Efficiency
Covariance Structure	Variance Components
Estimation Method	ML
Residual Variance Method	Profile
Fixed Effects SE Method	Model-Based
Degrees of Freedom Method	Satterthwaite

Class Level Information		
Class	Levels	Values
FactorA	3	1 2 3
FactorB	4	1 2 3 4

Dimensions	
Covariance Parameters	3
Columns in X	4
Columns in Z	16
Subjects	1
Max Obs per Subject	48

Number of Observations	
Number of Observations Read	48
Number of Observations Used	48
Number of Observations Not Used	0

Iteration History			
Iteration	Evaluations	-2 Log Like	Criterion
0	1	228.51318953	
1	2	211.65699758	0.00000281

Iteration History			
Iteration	Evaluations	-2 Log Like	Criterion
2	1	211.65682174	0.00000000

Convergence criteria met.

Estimated G matrix is not positive definite.

Covariance Parameter Estimates	
Cov Parm	Estimate
FactorB	2.8515
FactorA*FactorB	0
Residual	3.9882

Fit Statistics	
-2 Log Likelihood	211.7
AIC (Smaller is Better)	221.7
AICC (Smaller is Better)	223.1
BIC (Smaller is Better)	218.6

Type 3 Tests of Fixed Effects				
Effect	Num DF	Den DF	F Value	Pr > F
FactorA	2	44	18.85	<.0001

PROC GLM: Main Effects Tests (Unrestricted)

The GLM Procedure

Class Level Information		
Class	Levels	Values
FactorA	3	1 2 3
FactorB	4	1 2 3 4

Number of Observations Read	48
Number of Observations Used	48

PROC GLM: Main Effects Tests (Unrestricted)

The GLM Procedure

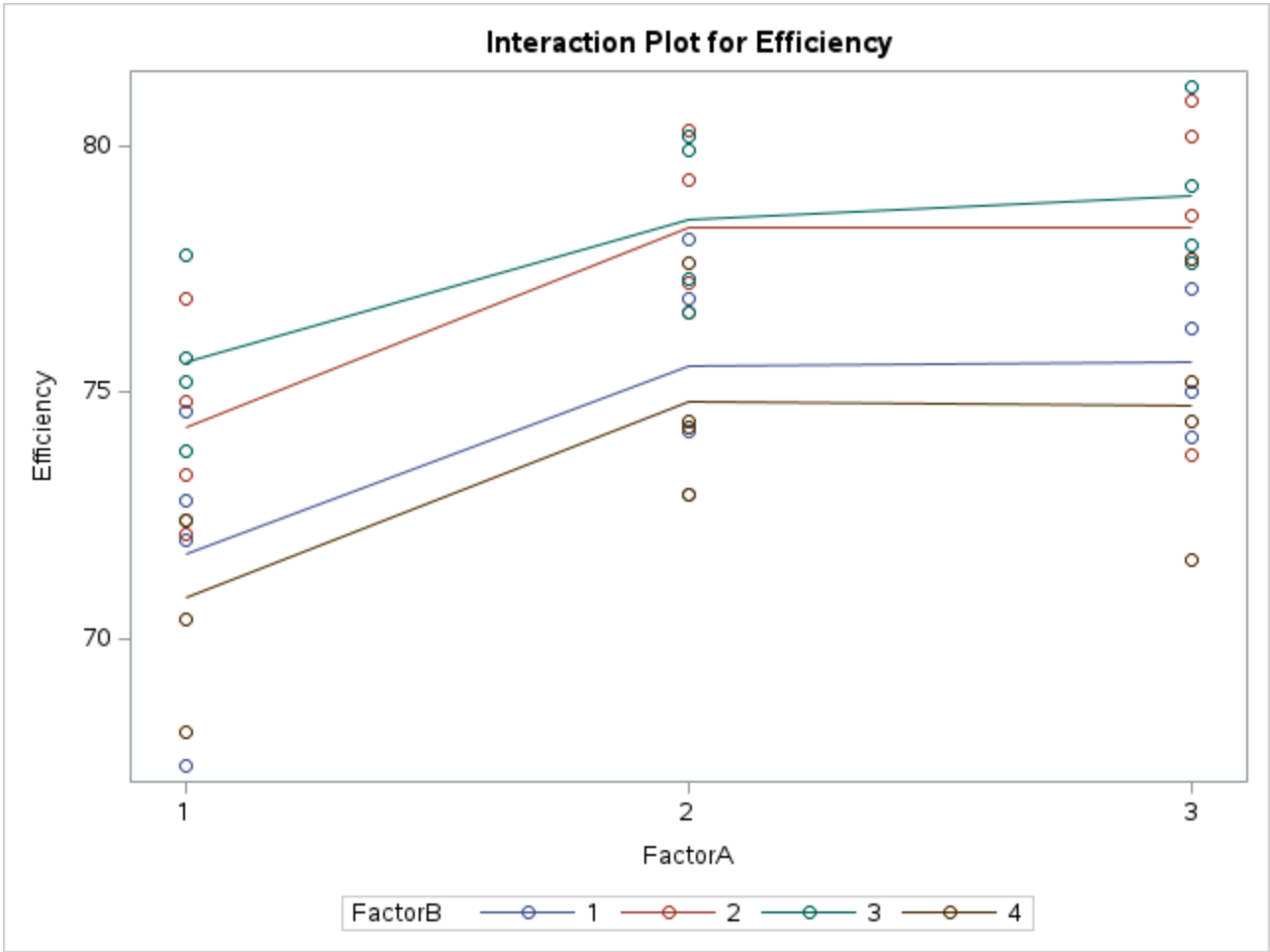
Dependent Variable: Efficiency

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	11	305.0916667	27.7356061	5.75	<.0001
Error	36	173.6250000	4.8229167		
Corrected Total	47	478.7166667			

R-Square	Coeff Var	Root MSE	Efficiency Mean
0.637312	2.904593	2.196114	75.60833

Source	DF	Type III SS	Mean Square	F Value	Pr > F
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Source	DF	Type III SS	Mean Square	F Value	Pr > F
FactorA	2	150.3879167	75.1939583	15.59	<.0001
FactorB	3	152.8516667	50.9505556	10.56	<.0001
FactorA*FactorB	6	1.8520833	0.3086806	0.06	0.9988



PROC GLM: Main Effects Tests (Unrestricted)

The GLM Procedure

Source	Type III Expected Mean Square
FactorA	Var(Error) + 4 Var(FactorA*FactorB) + Q(FactorA)
FactorB	Var(Error) + 4 Var(FactorA*FactorB) + 12 Var(FactorB)
FactorA*FactorB	Var(Error) + 4 Var(FactorA*FactorB)

PROC GLM: Main Effects Tests (Unrestricted)

The GLM Procedure
Tests of Hypotheses for Mixed Model Analysis of Variance

Dependent Variable: Efficiency

Source	DF	Type III SS	Mean Square	F Value	Pr > F
FactorA	2	150.387917	75.193958	243.60	<.0001
FactorB	3	152.851667	50.950556	165.06	<.0001
Error	6	1.852083	0.308681		

Source	DF	Type III SS	Mean Square	F Value	Pr > F
Error: MS(FactorA*FactorB)					

Source	DF	Type III SS	Mean Square	F Value	Pr > F
FactorA*FactorB	6	1.852083	0.308681	0.06	0.9988
Error: MS(Error)	36	173.625000	4.822917		

PROC MIXED: Fixed Effect A and Variance Components (REML)

The Mixed Procedure

Model Information	
Data Set	WORK.TRAINING
Dependent Variable	Efficiency
Covariance Structure	Variance Components
Estimation Method	REML
Residual Variance Method	Profile
Fixed Effects SE Method	Model-Based
Degrees of Freedom Method	Satterthwaite

Class Level Information		
Class	Levels	Values
FactorA	3	1 2 3
FactorB	4	1 2 3 4

Dimensions	
Covariance Parameters	3
Columns in X	4
Columns in Z	16
Subjects	1
Max Obs per Subject	48

Number of Observations	
Number of Observations Read	48
Number of Observations Used	48
Number of Observations Not Used	0

Iteration History			
Iteration	Evaluations	-2 Res Log Like	Criterion
0	1	225.45311480	
1	2	207.86808217	0.00000079
2	1	207.86803230	0.00000000

Convergence criteria met.

Estimated G matrix is not positive definite.

Covariance Parameter Estimates	
Cov Parm	Estimate
FactorB	3.8974

Covariance Parameter Estimates	
Cov Parm	Estimate
FactorA*FactorB	0
Residual	4.1780

Fit Statistics	
-2 Res Log Likelihood	207.9
AIC (Smaller is Better)	211.9
AICC (Smaller is Better)	212.2
BIC (Smaller is Better)	210.6

Type 3 Tests of Fixed Effects				
Effect	Num DF	Den DF	F Value	Pr > F
FactorA	2	42	18.00	<.0001

Comparison of Factor A Means (REML)

The Mixed Procedure

Model Information	
Data Set	WORK.TRAINING
Dependent Variable	Efficiency
Covariance Structure	Variance Components
Estimation Method	REML
Residual Variance Method	Profile
Fixed Effects SE Method	Model-Based
Degrees of Freedom Method	Satterthwaite

Class Level Information		
Class	Levels	Values
FactorA	3	1 2 3
FactorB	4	1 2 3 4

Dimensions	
Covariance Parameters	3
Columns in X	4
Columns in Z	16
Subjects	1
Max Obs per Subject	48

Number of Observations	
Number of Observations Read	48
Number of Observations Used	48
Number of Observations Not Used	0

Iteration History			
Iteration	Evaluations	-2 Res Log Like	Criterion
0	1	225.45311480	
1	2	207.86808217	0.00000079

Iteration History			
Iteration	Evaluations	-2 Res Log Like	Criterion
2	1	207.86803230	0.00000000

Convergence criteria met.

Estimated G matrix is not positive definite.

Covariance Parameter Estimates	
Cov Parm	Estimate
FactorB	3.8974
FactorA*FactorB	0
Residual	4.1780

Fit Statistics	
-2 Res Log Likelihood	207.9
AIC (Smaller is Better)	211.9
AICC (Smaller is Better)	212.2
BIC (Smaller is Better)	210.6

Type 3 Tests of Fixed Effects				
Effect	Num DF	Den DF	F Value	Pr > F
FactorA	2	42	18.00	<.0001

Least Squares Means						
Effect	FactorA	Estimate	Standard Error	DF	t Value	Pr > t
FactorA	1	73.1062	1.1115	4.06	65.77	<.0001
FactorA	2	76.7937	1.1115	4.06	69.09	<.0001
FactorA	3	76.9250	1.1115	4.06	69.21	<.0001

Differences of Least Squares Means									
Effect	FactorA	_FactorA	Estimate	Standard Error	DF	t Value	Pr > t	Adjustment	Adj P
FactorA	1	2	-3.6875	0.7227	42	-5.10	<.0001	Tukey-Kramer	<.0001
FactorA	1	3	-3.8187	0.7227	42	-5.28	<.0001	Tukey-Kramer	<.0001
FactorA	2	3	-0.1312	0.7227	42	-0.18	0.8568	Tukey-Kramer	0.9820

PROC MIXED (ML): Fixed Effect A and Variance Components

The Mixed Procedure

Model Information	
Data Set	WORK.TRAINING
Dependent Variable	Efficiency
Covariance Structure	Variance Components
Estimation Method	ML
Residual Variance Method	Profile
Fixed Effects SE Method	Model-Based
Degrees of Freedom Method	Satterthwaite

Class Level Information		
Class	Levels	Values
FactorA	3	1 2 3
FactorB	4	1 2 3 4

Dimensions	
Covariance Parameters	3
Columns in X	4
Columns in Z	16
Subjects	1
Max Obs per Subject	48

Number of Observations	
Number of Observations Read	48
Number of Observations Used	48
Number of Observations Not Used	0

Iteration History			
Iteration	Evaluations	-2 Log Like	Criterion
0	1	228.51318953	
1	2	211.65699758	0.00000281
2	1	211.65682174	0.00000000

Convergence criteria met.

Estimated G matrix is not positive definite.

Covariance Parameter Estimates	
Cov Parm	Estimate
FactorB	2.8515
FactorA*FactorB	0
Residual	3.9882

Fit Statistics	
-2 Log Likelihood	211.7
AIC (Smaller is Better)	221.7
AICC (Smaller is Better)	223.1
BIC (Smaller is Better)	218.6

Solution for Fixed Effects						
Effect	FactorA	Estimate	Standard Error	DF	t Value	Pr > t
Intercept		76.9250	0.9809	5.82	78.42	<.0001
FactorA	1	-3.8187	0.7061	44	-5.41	<.0001
FactorA	2	-0.1312	0.7061	44	-0.19	0.8534
FactorA	3	0	.	.	.	.

Type 3 Tests of Fixed Effects				
Effect	Num DF	Den DF	F Value	Pr > F
FactorA	2	44	18.85	<.0001

PROC MIXED (ML): Full Model for LRT H0: sigma2_B = 0**The Mixed Procedure**

Model Information	
Data Set	WORK.TRAINING
Dependent Variable	Efficiency
Covariance Structure	Variance Components
Estimation Method	ML
Residual Variance Method	Profile
Fixed Effects SE Method	Model-Based
Degrees of Freedom Method	Satterthwaite

Class Level Information		
Class	Levels	Values
FactorA	3	1 2 3
FactorB	4	1 2 3 4

Dimensions	
Covariance Parameters	3
Columns in X	4
Columns in Z	16
Subjects	1
Max Obs per Subject	48

Number of Observations	
Number of Observations Read	48
Number of Observations Used	48
Number of Observations Not Used	0

Iteration History			
Iteration	Evaluations	-2 Log Like	Criterion
0	1	228.51318953	
1	2	211.65699758	0.00000281
2	1	211.65682174	0.00000000

Convergence criteria met.

Estimated G matrix is not positive definite.

Covariance Parameter Estimates	
Cov Parm	Estimate
FactorB	2.8515
FactorA*FactorB	0
Residual	3.9882

Fit Statistics	
-2 Log Likelihood	211.7
AIC (Smaller is Better)	221.7
AICC (Smaller is Better)	223.1

Fit Statistics	
BIC (Smaller is Better)	218.6

Solution for Fixed Effects						
Effect	FactorA	Estimate	Standard Error	DF	t Value	Pr > t
Intercept		76.9250	0.9809	5.82	78.42	<.0001
FactorA	1	-3.8187	0.7061	44	-5.41	<.0001
FactorA	2	-0.1312	0.7061	44	-0.19	0.8534
FactorA	3	0	.	.	.	.

Type 3 Tests of Fixed Effects				
Effect	Num DF	Den DF	F Value	Pr > F
FactorA	2	44	18.85	<.0001

PROC MIXED (ML): Reduced Model for LRT H0: sigma2_B = 0

The Mixed Procedure

Model Information	
Data Set	WORK.TRAINING
Dependent Variable	Efficiency
Covariance Structure	Variance Components
Estimation Method	ML
Residual Variance Method	Profile
Fixed Effects SE Method	Model-Based
Degrees of Freedom Method	Satterthwaite

Class Level Information		
Class	Levels	Values
FactorA	3	1 2 3
FactorB	4	1 2 3 4

Dimensions	
Covariance Parameters	2
Columns in X	4
Columns in Z	12
Subjects	1
Max Obs per Subject	48

Number of Observations	
Number of Observations Read	48
Number of Observations Used	48
Number of Observations Not Used	0

Iteration History			
Iteration	Evaluations	-2 Log Like	Criterion
0	1	228.51318953	
1	1	223.53900242	0.00000000

Convergence criteria met.

Covariance Parameter Estimates	
Cov Parm	Estimate
FactorA*FactorB	2.0173
Residual	4.8229

Fit Statistics	
-2 Log Likelihood	223.5
AIC (Smaller is Better)	233.5
AICC (Smaller is Better)	235.0
BIC (Smaller is Better)	236.0

Solution for Fixed Effects						
Effect	FactorA	Estimate	Standard Error	DF	t Value	Pr > t
Intercept		76.9250	0.8976	12	85.70	<.0001
FactorA	1	-3.8187	1.2694	12	-3.01	0.0109
FactorA	2	-0.1312	1.2694	12	-0.10	0.9194
FactorA	3	0	.	.	.	.

Type 3 Tests of Fixed Effects				
Effect	Num DF	Den DF	F Value	Pr > F
FactorA	2	12	5.83	0.0170

PROC MIXED (ML): CI for Variance Components

The Mixed Procedure

Model Information	
Data Set	WORK.TRAINING
Dependent Variable	Efficiency
Covariance Structure	Variance Components
Estimation Method	ML
Residual Variance Method	Profile
Fixed Effects SE Method	Model-Based
Degrees of Freedom Method	Satterthwaite

Class Level Information		
Class	Levels	Values
FactorA	3	1 2 3
FactorB	4	1 2 3 4

Dimensions	
Covariance Parameters	3
Columns in X	4
Columns in Z	16
Subjects	1
Max Obs per Subject	48

Number of Observations	
Number of Observations Read	48
Number of Observations Used	48
Number of Observations Not Used	0

Iteration History			
Iteration	Evaluations	-2 Log Like	Criterion
0	1	228.51318953	
1	2	211.65699758	0.00000281
2	1	211.65682174	0.00000000

Convergence criteria met.

Estimated G matrix is not positive definite.

Covariance Parameter Estimates				
Cov Parm	Estimate	Alpha	Lower	Upper
FactorB	2.8515	0.05	0.9397	34.6773
FactorA*FactorB	0	.	.	.
Residual	3.9882	0.05	2.7333	6.3638

Fit Statistics	
-2 Log Likelihood	211.7
AIC (Smaller is Better)	221.7
AICC (Smaller is Better)	223.1
BIC (Smaller is Better)	218.6

Solution for Fixed Effects						
Effect	FactorA	Estimate	Standard Error	DF	t Value	Pr > t
Intercept		76.9250	0.9809	5.82	78.42	<.0001
FactorA	1	-3.8187	0.7061	44	-5.41	<.0001
FactorA	2	-0.1312	0.7061	44	-0.19	0.8534
FactorA	3	0	.	.	.	.

Type 3 Tests of Fixed Effects				
Effect	Num DF	Den DF	F Value	Pr > F
FactorA	2	44	18.85	<.0001

PROC MIXED (ML): Factor A Means Comparison

The Mixed Procedure

Model Information	
Data Set	WORK.TRAINING
Dependent Variable	Efficiency
Covariance Structure	Variance Components
Estimation Method	ML
Residual Variance Method	Profile
Fixed Effects SE Method	Model-Based

Model Information	
Degrees of Freedom Method	Satterthwaite

Class Level Information		
Class	Levels	Values
FactorA	3	1 2 3
FactorB	4	1 2 3 4

Dimensions	
Covariance Parameters	3
Columns in X	4
Columns in Z	16
Subjects	1
Max Obs per Subject	48

Number of Observations	
Number of Observations Read	48
Number of Observations Used	48
Number of Observations Not Used	0

Iteration History			
Iteration	Evaluations	-2 Log Like	Criterion
0	1	228.51318953	
1	2	211.65699758	0.00000281
2	1	211.65682174	0.00000000

Convergence criteria met.

Estimated G matrix is not positive definite.

Covariance Parameter Estimates	
Cov Parm	Estimate
FactorB	2.8515
FactorA*FactorB	0
Residual	3.9882

Fit Statistics	
-2 Log Likelihood	211.7
AIC (Smaller is Better)	221.7
AICC (Smaller is Better)	223.1
BIC (Smaller is Better)	218.6

Solution for Fixed Effects						
Effect	FactorA	Estimate	Standard Error	DF	t Value	Pr > t
Intercept		76.9250	0.9809	5.82	78.42	<.0001
FactorA	1	-3.8187	0.7061	44	-5.41	<.0001
FactorA	2	-0.1312	0.7061	44	-0.19	0.8534
FactorA	3	0	.	.	.	.

Type 3 Tests of Fixed Effects				
Effect	Num DF	Den DF	F Value	Pr > F
FactorA	2	44	18.85	<.0001

Least Squares Means						
Effect	FactorA	Estimate	Standard Error	DF	t Value	Pr > t
FactorA	1	73.1062	0.9809	5.82	74.53	<.0001
FactorA	2	76.7937	0.9809	5.82	78.29	<.0001
FactorA	3	76.9250	0.9809	5.82	78.42	<.0001

Differences of Least Squares Means									
Effect	FactorA	_FactorA	Estimate	Standard Error	DF	t Value	Pr > t	Adjustment	Adj P
FactorA	1	2	-3.6875	0.7061	44	-5.22	<.0001	Tukey-Kramer	<.0001
FactorA	1	3	-3.8187	0.7061	44	-5.41	<.0001	Tukey-Kramer	<.0001
FactorA	2	3	-0.1312	0.7061	44	-0.19	0.8534	Tukey-Kramer	0.9811